

Sepid (Sepidehsadat) Hosseini

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Education

Simon Fraser University

PhD in Computing Science, GPA: 4.11/4.3

British Columbia, Canada

May 2025

- Field: 3D Computer Vision and Computer Graphics.

Seoul National University

Master of Science in Electrical and Computer Engineering, GPA: 95.1/100

Seoul, South Korea

August 2019

- Field: Intelligent Signal Processing, Computer Vision, Facial and Gesture Recognition.

Seoul National University

Bachelor of Science in Electrical and Computer Engineering, GPA: 88.6/100

Seoul, South Korea

March 2017

- Field: Smart Road Implementation Using Arduino Car.

Work Experience

RBC Borealis (Borealis AI)

Senior Machine Learning Researcher / Tech Lead

Vancouver, Canada

May 2026–Present

- Technical lead for ATOM, the Asynchronous Temporal Model, a foundation model for financial services focused on fraud detection, anomaly detection, and personalization.
- Leading research-to-production delivery of large-scale temporal modeling solutions in collaboration with research, engineering, and product teams.

RBC Borealis (Borealis AI)

Machine Learning Researcher

Vancouver, Canada

Dec 2023–May 2026

- Core researcher on the fraud detection team, contributing to ATOM and its adaptation to recommender systems and personalized financial applications.
- Conducted applied machine learning research leading to publications, patents, and production-oriented model development.

RBC Borealis (Borealis AI)

Machine Learning Research Intern

Vancouver, Canada

May 2023–Sep 2023

- Conducted research on prompt-based learning for efficient temporal domain generalization under distribution shift.

LG Electronics, CTO, Intelligent Systems Research Center

Research Intern

Seoul, Korea

Jul 2017–Sep 2017

- Developed capsule-network-based methods for face spoofing detection and robust facial analysis.

Samsung Techwin (Hanwha)

Software Intern

Seoul, Korea

Jul 2015–Aug 2015

- Worked on object tracking algorithms for video analysis and surveillance-related applications.

Selected Publications

FairNVT: Fair Classification via Noise Injection in Vision Transformers

TMLR 2026

Qiaoyue Tang, Sepidehsadat Hosseini, Mengyao Zhai, Thibaut Durand, Greg Mori

Scalable Temporal Domain Generalization via Prompting

PUT Workshop, ICML 2025

Sepidehsadat Hosseini, Mengyao Zhai, Hossein Hajimirsadeghi, Frederick Tung

Radar: Fast Long-Context Decoding for Any Transformer

ICLR 2025

Yongchang Hao, Mengyao Zhai, Hossein Hajimirsadeghi, Sepidehsadat Hosseini, Fred Tung

PuzzleFusion: Unleashing the Power of Diffusion Models for Spatial Puzzle Solving

NeurIPS 2023 Spotlight

Sepidehsadat Hosseini, Mohammad Amin Shabani, Saghar Irandoust, Yasutaka Furukawa

Floorplan Restoration by Structure Hallucinating Transformer Cascades

BMVC 2023

Sepidehsadat Hosseini, Yasutaka Furukawa

HouseDiffusion: Vector Floorplan Generation via a Diffusion Model with Discrete and Continuous Denoising

CVPR 2023

Mohammad Amin Shabani, **Sepidehsadat Hosseini**, Yasutaka Furukawa

House-GAN++: Generative Adversarial Layout Refinement Network

CVPR 2021

Nelson Nauata, **Sepidehsadat Hosseini**, Kai-Hung Chang, Hang Chu, Chin-Yi Cheng, Yasutaka Furukawa

GF-CapsNet: Using Gabor Jet and Capsule Networks for Facial Age, Gender, and Expression Recognition

FG 2019

Sepidehsadat Hosseini, Nam Ik Cho

Patents

Neural Network Temporal Domain Generalization Method and System

US App. 18/891,940

Sepidehsadat Hosseini, Mengyao Zhai, Hossein Hajimirsadeghi, Frederick Tung. Assignee: Royal Bank of Canada. Filed 2024.

Application of Representation Learning to Discrete Event Sequences Across Discrete Domains

PCT/CA2025/050052

Mohamed Osama Ahmed, Taivanbat Badamdorj, Ruizhi Deng, Peter Forsyth, **Sepidehsadat Hosseini**, Greg Mori, Gabriel L. Oliveira, Edward Smith, Frederick Tung, Ankit Vani, He Zhao. Assignee: Royal Bank of Canada. Filed 2025.

Memberships and Volunteering Experience

Catch, Adapt, and Operate: Monitoring ML Models Under Drift

Rio de Janeiro, Brazil

Lead Organizer and Program Chair

ICLR 2026

- Led the organization and program committee for an ICLR workshop on monitoring, adapting, and operating machine learning models under distribution shift.

LG Global Challenger

Seoul, Korea

Participant, Student Ambassador

2016–2017

- Selected for a fellowship program to travel across South Korea and promote Korean culture and innovation.

Seoul National University Dormitory

Seoul, Korea

Student Advisor

2013–2014

- Supported international and domestic students in residential life and community-building activities.

Bean Seoul

Seoul, Korea

Volunteer Instructor

2013–2017

- Volunteered in educational and recreational activities with children in orphanages.

IUST Robotics Team

Tehran, Iran

Team Member

2009–2012

- Participated in robotics competitions as part of the university robotics research team.

Honors and Awards

SFU Computing Science Graduate Fellowship

Fall 2019, Fall 2020, Fall 2021, Summer 2022, Fall 2022

SFU Computing Science Entrance Scholarship

Fall 2019

- Award valued at CAD 10,000.

Seoul National University Excellent Student Global Scholarship

2017, 2018

- Award valued at USD 8,000 per year.

Brain Korea Scholarship, Excellent Researcher Award

2017, 2018

LG International Student Ambassador

2016–2017

Korean Government Scholarship

2012–2017

- Award valued at approximately USD 97,000.